

SE LAB-1

Hostel Maintenance & Issue Ticketing System

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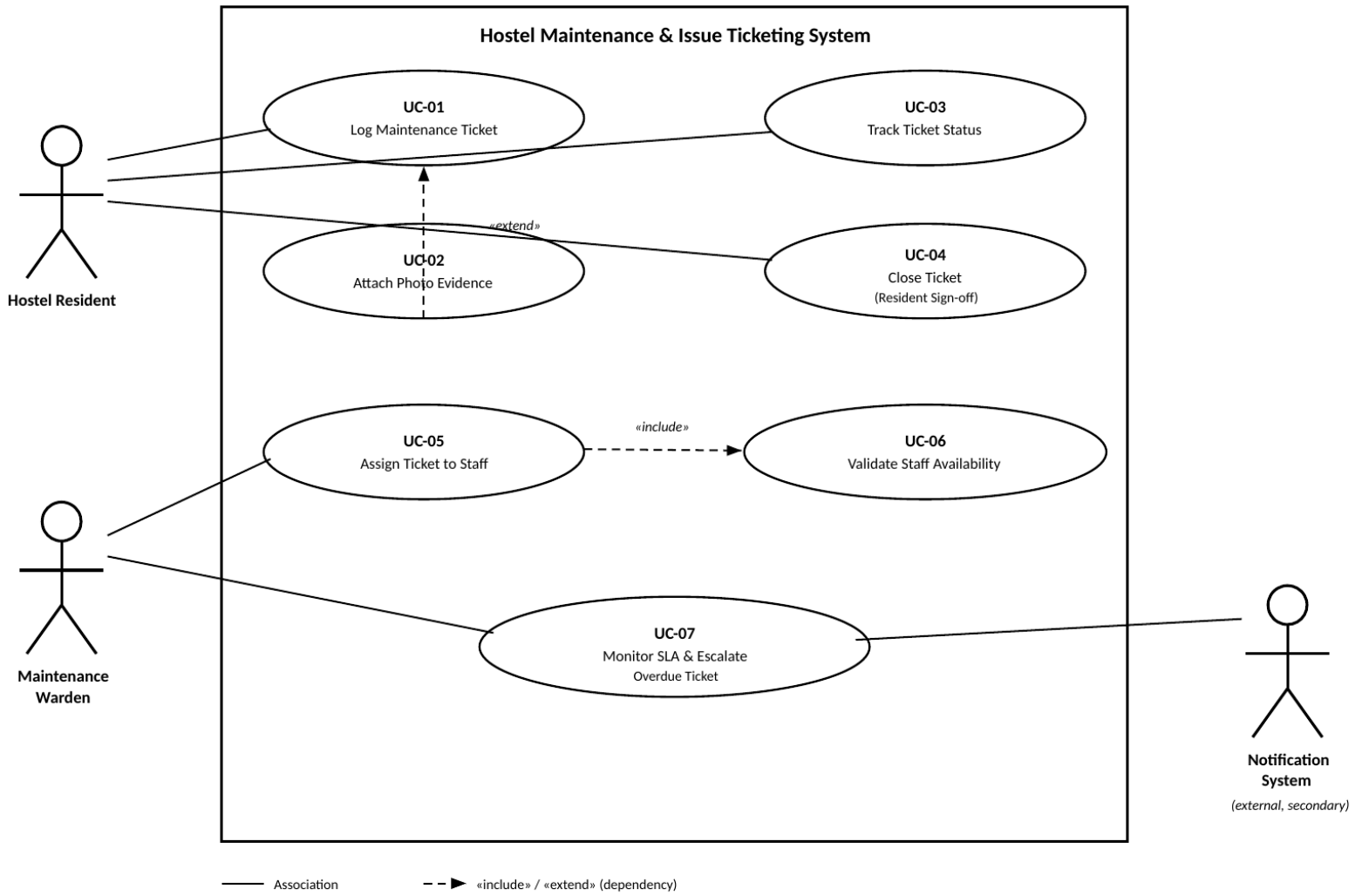
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Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: Ticket is created, assigned a unique ID, and routed to warden queue. Fail: Incomplete form accepted without room tag.	Given requirement — core entry point for all downstream ticketing workflow.
FR-002	Functional	The system shall allow the Maintenance Warden to assign a logged ticket to an available maintenance staff member filtered by issue category.	High	Pass: Warden selects staff from a category-filtered list and ticket status changes to "Assigned" with staff name and timestamp recorded. Fail: Ticket remains unassigned after 10 minutes, or staff outside the matching category can be selected.	Ensures the right skillset is routed to each issue and prevents misassignment.
FR-003	Functional	The system shall allow residents to view the real-time status of their submitted tickets (Open, Assigned, In Progress, Resolved, Closed).	Medium	Pass: Resident's ticket list reflects the current status within 5 seconds of a warden update. Fail: Displayed status is stale by more than 5 seconds or unavailable.	Improves transparency and reduces repeat status-check queries to the hostel office.
FR-004	Functional	The system shall require the reporting resident to provide a closure sign-off (confirmation and optional feedback) before a ticket can be marked Closed.	High	Pass: Ticket cannot transition to "Closed" without a recorded resident confirmation and timestamp. Fail: Ticket reaches "Closed" state without resident action.	Validates that the issue is actually resolved from the resident's perspective, not just the staff's.
FR-005	Functional	The system shall provide the Maintenance Warden a dashboard of open tickets filterable by category, severity, and SLA status.	Medium	Pass: Applying a filter returns only tickets matching all selected criteria. Fail: Filtered results include tickets outside the selected criteria.	Enables the warden to prioritize urgent tickets and balance staff workload.
NFR-001	Nonfunctional	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without a status change.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load.	Given requirement — protects against unresolved high-priority issues going unnoticed.
NFR-002	Nonfunctional	The system shall ensure ticket submission, including a compressed photo attachment, completes in under 5 seconds on a standard 4G connection.	Medium	Pass: Load testing confirms 95% of ticket submissions (with one photo) complete in under 5 seconds on simulated 4G. Fail: More than 5% of attempts exceed 5 seconds or fail to upload.	Keeps reporting fast and usable during urgent situations (e.g., a leak) when residents need to log the issue quickly.

UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case:

UC-01 — Log Maintenance Ticket

Primary Actor:

Hostel Resident

Secondary Actors:

Maintenance Warden (receives the routed ticket)

Preconditions:

- The resident is logged into the hostel ticketing system/app with a verified hostel room account.
- The resident is physically able to identify the issue category (plumbing, electrical, or carpentry) and the affected room number.

Postconditions:

- A new ticket exists in the system with a unique Ticket ID, status "Open", and is visible in the Maintenance Warden's queue.
- The resident can view the ticket and its status from their own ticket list.

Main Success Scenario:

1. Resident selects "Report an Issue" from the home screen.
2. System displays a ticket form requesting room number, issue category (Plumbing / Electrical / Carpentry), a short description, and severity (Low / Medium / High).
3. Resident enters the room number and selects an issue category and severity.
4. Resident optionally attaches a photo of the issue (see UC-02 — Attach Photo Evidence, which extends this use case).
5. Resident submits the ticket.
6. System validates that all required fields (room number, category, severity) are complete.
7. System generates a unique Ticket ID and geo-tags the ticket to the resident's registered room location.
8. System sets the ticket status to "Open" and routes it to the Maintenance Warden's queue, filtered by category.
9. System displays a confirmation screen to the resident showing the Ticket ID and expected SLA window.
10. Use case ends successfully.

Alternate Flow — 6a. Incomplete Form Submitted:

- 6a1. System detects a missing required field (e.g., no room number tag).
- 6a2. System rejects the submission and highlights the missing field(s) with an inline error message.
- 6a3. Resident corrects the form and resubmits, returning to step 6.
- 6a4. If the resident abandons the form without resubmitting, no ticket is created and the use case ends without success.

4. Requirements Traceability Matrix

Requirement	Use Case / Model Element	Purpose
FR-001	UC-01 Log Maintenance Ticket	Resident reports a maintenance issue.
FR-002	UC-04 Assign Maintenance Staff	Warden dispatches maintenance staff.
FR-003	UC-05 Update Ticket Status	Assigned staff records progress.
FR-004	UC-07 Confirm Issue Closure	Resident records closure sign-off.
FR-005	UC-08 Monitor SLA	Warden monitors maintenance deadlines.
NFR-001	UC-09 Escalate Overdue Ticket	Conditional escalation for overdue high-priority tickets.
NFR-002	Cross-cutting security constraint	Applies to authenticated/authorized system interactions.